

Software Requirements Specification (SRS)

Task Manager Application

Project: Task Manager Application

Version: 1.0

Authors:

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Revision History

Version	Date	Author	Change summary	Approval
0.1	2/09/2026	Arya M G	Initial outline and Introduction (Section 1) drafted	
0.2	2/09/2026	Aryan Burman	Overall Description and External Interface Requirements (Sections 2–3) added	
0.3	3/09/2026	Arushi Punhani	System Features and Non-Functional Requirements (Sections 4–5) added	
0.4	3/09/2026	Arnav Sharma	Quality Attributes, System Models, and RTM (Sections 6–8) added, formatting reviewed	

Approvals

Role	Name	Signature / Email	Date
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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Task Manager Application. It defines the functional and non-functional requirements, interfaces, and verification criteria for a web/mobile-based application that allows individuals and small teams to create, organize, track, and collaborate on tasks.

1.2 Scope

The Task Manager Application covers user registration and authentication, task creation and lifecycle management, categorization and search, task assignment and collaboration among registered users, reminders and notifications, and basic progress reporting. It excludes integration with third-party calendar/email providers and payment/billing functionality, which are considered out of scope for the current version.

1.3 Audience

Developers, QA Engineers, Project Reviewers/Instructors, and End Users evaluating the system's requirements.

1.4 Definitions

List of acronyms: UI (User Interface), API (Application Programming Interface), CRUD (Create, Read, Update, Delete), JWT (JSON Web Token), RBAC (Role-Based Access Control), TLS (Transport Layer Security), NFR (Non-Functional Requirement), RTM (Requirements Traceability Matrix), WCAG (Web Content Accessibility Guidelines).

2. Overall Description

2.1 Product Perspective

The Task Manager Application is a standalone, self-contained web/mobile application with a client-side UI, a backend API, and a persistent database. It is not dependent on any legacy system but may integrate with an external notification/email service for reminders.

2.2 Major Product Functions

- User registration, login, and password recovery
- Task creation, editing, deletion, and completion tracking
- Categorization and tagging of tasks
- Search, filter, and sort of tasks
- Task assignment and sharing among users
- Comments/collaboration on shared tasks
- Due-date and assignment notifications
- Progress dashboard and report export
- Administrative user management and audit logging

2.3 User Roles and Characteristics

- End User: registers, creates and manages their own tasks; may share/assign tasks to teammates.
- Teammate/Collaborator: an end user granted access to another user's shared task or list.
- System Administrator: manages user accounts, monitors system health, and reviews audit logs.

2.4 Operating Environment

Modern web browsers (Chrome, Firefox, Edge, Safari) on desktop and mobile; backend deployed on a cloud application server with a relational or document database; internet connectivity required for real-time sync and notifications.

2.5 Constraints

Must use secure password storage and TLS for all traffic; database schema must support multi-user sharing/permissions; notification delivery depends on an external email/push provider's availability.

3. External Interface Requirements

3.1 User Interfaces

A responsive web UI (and optionally a mobile app) with a task list/board view, task detail view, and dashboard. Accessibility features include screen-reader friendly labels and high-contrast mode support.

3.2 Hardware Interfaces

None required beyond a standard client device (desktop, laptop, tablet, or smartphone) with an internet connection; no specialized hardware is used.

3.3 Software Interfaces

- Backend REST API (JSON over HTTPS) for all client-server communication.
- Database (relational or NoSQL) for persistent storage of users, tasks, and audit logs.
- External Notification/Email Service API for reminders and assignment alerts.

3.4 Communications

TLS 1.2+ is enforced for all client-server and server-to-service communication. The client uses token-based (JWT) authentication on each API request after login.

4. System Features (Detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow TM-F-####. A total of 20 functional requirements are specified across the feature areas below.

4.1 Authentication & Account Management

Description: Allow users to register, authenticate, recover access, and protect accounts from unauthorized access.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance Criteria / Test Case Ref	Comments/ Dependencies

TM-F-001	The system shall allow a new user to register using a valid email address and password.	Functional	High	Product Owner	AC-TM-F-001: Valid email + password (min 8 chars) creates an account and sends a verification email. Test: TC-Auth-01	Requires email service
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Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance Criteria / Test Case Ref	Comments/ Dependencies
TM-F-002	The system shall authenticate a registered user via email and password before granting access to tasks.	Functional	High	Security	AC-TM-F-002: Correct credentials return a valid session token; incorrect ones are rejected. Test: TC-Auth-02	Depends on TM-F-001
TM-F-003	The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts and log the event.	Functional	High	Security	AC-TM-F-003: 5th failure locks account and logs an audit entry. Test: TC-Auth-03	Audit log required
TM-F-004	The system shall allow a user to reset a forgotten password via a time limited email link.	Functional	Medium	Support Team	AC-TM-F-004: Link expires after 30 minutes; used link cannot be reused. Test: TC-Auth-04	Email service

4.2 Task Creation & Lifecycle Management

Description: Core CRUD operations on tasks, including status transitions, organization, search, and sorting.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance Criteria / Test Case Ref	Comments/ Dependencies
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TM-F 010	The system shall allow an authenticated user to create a task with a title, description, due date, and priority.	Functional	High	End User	AC-TM-F-010: Task with required fields is saved and visible in the task list. Test: TC-Task-01	Requires TM-F 002
TM-F 011	The system shall allow a user to edit the title, description, due date, priority, or status of an existing task they own.	Functional	High	End User	AC-TM-F-011: Edited fields persist and are reflected immediately in the UI. Test: TC-Task-02	Ownership check required
TM-F 012	The system shall allow a user to delete a task they own, with a confirmation prompt before permanent removal.	Functional	Medium	End User	AC-TM-F-012: Task is removed from list only after confirmation; deletion is logged. Test: TC-Task-03	Soft delete recommended

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance Criteria / Test Case Ref	Comments/ Dependencies
TM-F 013	The system shall allow a user to mark a task as complete or reopen a completed task.	Functional	High	End User	AC-TM-F-013: Status toggle updates task state and completion timestamp. Test: TC Task-04	-
TM-F 014	The system shall allow a user to organize tasks using categories or custom tags.	Functional	Medium	End User	AC-TM-F-014: Tasks can be filtered by an assigned category/tag. Test: TC-Task-05	-
TM-F 015	The system shall allow a user to search and filter tasks by title, status, due date, priority, or tag.	Functional	Medium	End User	AC-TM-F-015: Search returns only tasks matching the given filter criteria. Test: TC-Task-06	-

TM-F-016	The system shall allow a user to sort the task list by due date, priority, or creation date.	Functional	Low	End User	AC-TM-F-016: Sort order updates the displayed list without a full page reload. Test: TC-Task-07	-
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4.3 Collaboration & Sharing

Description: Allow tasks and task lists to be shared or assigned across registered users, with comments.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance Criteria / Test Case Ref	Comments/ Dependencies
TM-F-020	The system shall allow a user to assign a task to another registered user (teammate).	Functional	High	Product Owner	AC-TM-F-020: Assigned task appears in the teammate's task list. Test: TC-Collab-01	Requires multi user accounts
TM-F-021	The system shall allow users to add comments to a shared or assigned task.	Functional	Medium	End User	AC-TM-F-021: Comment is timestamped, attributed to the author, and visible to all task collaborators. Test: TC-Collab-02	-

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance Criteria / Test Case Ref	Comments/ Dependencies
TM-F-022	The system shall allow a user to share a task list with one or more teammates with view or edit permission.	Functional	Medium	End User	AC-TM-F-022: Shared users can access the list according to the granted permission level. Test: TC Collab-03	Permission model required

4.4 Notifications & Reminders

Description: Notify users of upcoming due dates and new task assignments.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance Criteria / Test Case Ref	Comments/ Dependencies
TM-F-030	The system shall send a reminder notification to the task owner a configurable time before the due date.	Functional	Medium	End User	AC-TM-F-030: Notification is delivered within 1 minute of the scheduled reminder time. Test: TC-Notif 01	Notification service
TM-F-031	The system shall send a notification to a teammate when a task is assigned to them.	Functional	Medium	End User	AC-TM-F-031: Notification is delivered immediately after assignment. Test: TC-Notif-02	Depends on TM-F-020

4.5 Reporting & Dashboard

Description: Provide a summarized view of task status and allow export of task data.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance Criteria / Test Case Ref	Comments/ Dependencies
TM-F-040	The system shall display a dashboard summarizing task counts by status (pending, in progress, completed, overdue).	Functional	Medium	Product Owner	AC-TM-F-040: Dashboard counts match the underlying task data. Test: TC Report-01	-

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance Criteria / Test Case Ref	Comments/ Dep
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						end enci es
TM-F 041	The system shall allow a user to export their task list as a CSV or PDF report.	Functional	Low	End User	AC-TM-F-041: Exported file contains all visible tasks with correct field values. Test: TC-Report-02	-

4.6 Administration & Maintenance

Description: Provide administrators the ability to manage user accounts and review audit trails.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance Criteria / Test Case Ref	Com me nts/ Dep end enci es
TM-F 050	The system shall provide an admin interface to view, disable, or delete user accounts.	Functional	High	System Admin	AC-TM-F-050: Admin actions on a user account take effect immediately and are logged. Test: TC-Admin-01	Admin role required
TM-F 051	The system shall log all administrative actions with timestamp, admin identity, and action type.	Functional	High	Security/ Audit	AC-TM-F-051: Every admin action produces a corresponding audit log entry. Test: TC Admin-02	-

5. Non-Functional Requirements (Detailed)

NFRs below are measurable and tied to test plans. IDs follow TM-NF-###.

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
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TM-NF 001	The system shall return task list views within 2 seconds for 95% of requests under normal load (up to 500 concurrent users).	Performance	High	95th percentile response time ≤ 2s in load test. Test: TC-Perf-01
TM-NF 002	The system shall provide 99.5% monthly uptime, excluding scheduled maintenance windows.	Reliability	High	Uptime reports show ≥99.5% per month. Test: Ops monitoring report

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
TM-NF 003	The system shall support at least 10,000 registered users and 1,000,000 tasks without degradation in response time.	Scalability	Medium	Load test at target scale meets TM-NF 001 thresholds. Test: TC-Scale-01
TM-NF 004	The user interface shall conform to WCAG 2.1 AA accessibility guidelines, including screen-reader support and keyboard navigation.	Usability/ Accessibility	Medium	Accessibility audit pass. Test: TC-UX 01
TM-NF 005	The system shall retain deleted task records in a recoverable state for 30 days before permanent purge.	Data Retention	Medium	Deleted task is recoverable within 30 days; purged after. Test: TC-Data-01
TM-NF 006	The system shall be usable on both desktop browsers and mobile devices via a responsive interface.	Usability	Medium	UI renders correctly on defined breakpoints (≥360px width). Test: TC UX-02

5.1 Security

5.1.1 Security Objectives

- Protect user credentials and personal data from unauthorized access, disclosure, or modification.
- Ensure that only authenticated, authorized users can view, create, edit, or delete their own tasks and data.
- Maintain an auditable record of security-relevant events (logins, failed attempts, admin actions) to support incident investigation.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria / Test Case Ref
TM-SR 001	The system shall enforce TLS 1.2 or higher for all client-server network communications.	Security	High	AC-TM-SR-001: Connections on plain HTTP are rejected/redirected. Test: TC-Sec-01
TM-SR 002	The system shall store user passwords using a salted, industry-standard hashing algorithm (e.g., bcrypt/Argon2); plaintext passwords shall never be stored.	Security	High	AC-TM-SR-002: Database inspection shows no plaintext passwords. Test: TC Sec-02

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria / Test Case Ref
TM-SR 003	The system shall enforce role-based access control so a user can only access, edit, or delete tasks they own or have been explicitly granted access to.	Security	High	AC-TM-SR-003: Attempt to access another user's private task returns Forbidden/403. Test: TC-Sec-03
TM-SR 004	The system shall issue and validate session tokens (e.g., JWT) with a defined expiry, and invalidate tokens on logout.	Security	High	AC-TM-SR-004: Expired or invalidated tokens are rejected on subsequent requests. Test: TC-Sec-04
TM-SR 005	The system shall log and alert on repeated failed authentication attempts to detect potential brute force activity.	Security	Medium	AC-TM-SR-005: 5+ failed attempts within 10 minutes generate a security alert. Test: TC-Sec-05

6. Quality Attributes & Acceptance Tests

Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and the RTM shows all listed test cases passed.

Acceptance test suites: Authentication, Task Management, Collaboration, Notifications, Reporting, Performance, Security, and Accessibility.

7. System Models and Diagrams

7.1 UML Use-Case Diagrams

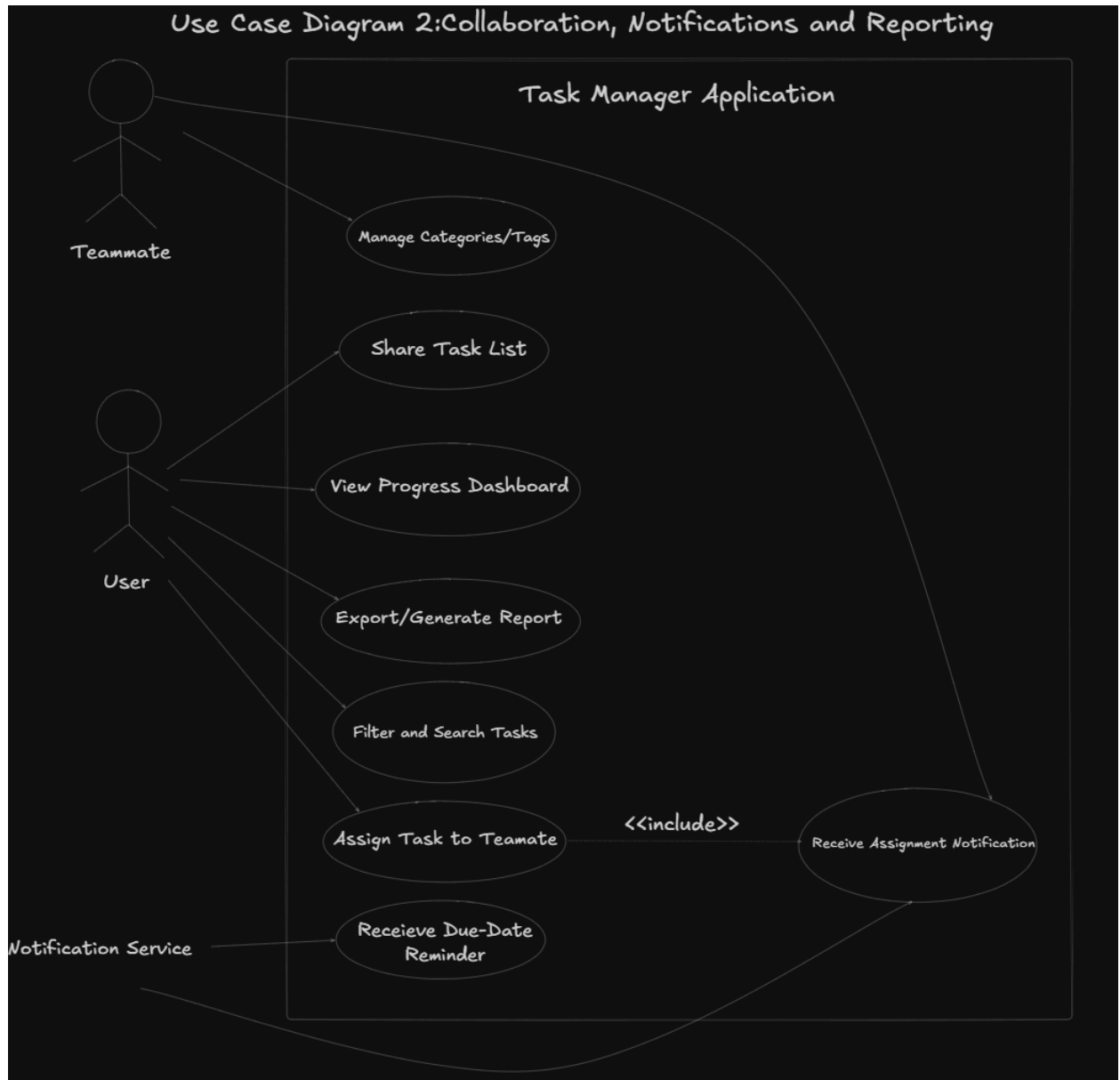
At least 2 UML use-case diagrams are to be inserted here, covering:

- Core task management & authentication (actors: User, Admin)
- Collaboration, notifications & reporting (actors: User, Teammate, Notification Service)

Use Case Diagram 1: Core Task Management & Authentication



Use Case Diagram 2: Collaboration, Notifications and Reporting



8. Requirements Traceability Matrix (RTM)

The RTM below maps a representative subset of requirements to design specifications, implementation modules, and test cases. This matrix should be expanded to cover all requirements as design and test artifacts are finalized.

Req ID	Requirement (short)	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
TM-F-001	User registration	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	-
TM-F-002	User login	4.1 / DS-Auth-02	AuthModule	TC-Auth-02	N	-
TM-F-010	Create task	4.2 / DS-Task-01	TaskModule	TC-Task-01	N	-
TM-F-011	Edit task	4.2 / DS-Task-02	TaskModule	TC-Task-02	N	-
TM-F-013	Mark task complete	4.2 / DS-Task-04	TaskModule	TC-Task-04	N	-
TM-F-020	Assign task	4.3 / DS-Collab 01	CollabModule	TC-Collab-01	N	-
TM-F-030	Due-date reminder	4.4 / DS-Notif-01	NotifModule	TC-Notif-01	N	Depends on Scheduler
TM-F-040	Dashboard summary	4.5 / DS-Report 01	ReportModule	TC-Report-01	N	-
TM-F-050	Admin user management	4.6 / DS-Admin 01	AdminModule	TC-Admin-01	N	-
TM-NF 001	Response time target	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N	-
TM-NF 002	Uptime target	5 / DS-Ops-01	Infra	Ops reports	N	-
TM-SR 002	Password hashing	5.1.2 / DS-Sec 02	AuthModule	TC-Sec-02	N	-
TM-SR 003	Role-based access control	5.1.2 / DS-Sec 03	AuthModule / TaskModule	TC-Sec-03	N	-